Limits

$$\lim_{x \to \infty} \left((x+1) - \sqrt{x^2+2x+4} \right)$$

$$\Rightarrow \frac{\left((x+1) - \sqrt{x^2+2x+4}\right)\left((x+1) + (\sqrt{x^2+2x+4})\right)}{\left((x+1) + \sqrt{x^2+2x+4}\right)}$$

$$\Rightarrow \frac{(x+1)^2 - \left(\sqrt{x^2+2x+4}\right)^2}{(x+1) + \sqrt{x^2+2x+4}}$$

$$\Rightarrow \frac{x^2+2x+1 - x^2-2x-4}{(x+1) + \sqrt{x^2+2x+4}} = \frac{-3}{(x+1) + \sqrt{x^2+2x+4}}$$

$$\Rightarrow \frac{-3}{\infty} = 0$$

Day 9